

NUST

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FLOOD MAPPING AND ASSESSMENT REPORT

Narowal District, Punjab, Pakistan

EXECUTIVE SUMMARY

This report presents flood mapping analysis of Narowal district using Sentinel-1 SAR imagery. Change detection between pre-flood (April 29, 2025) and post-flood (September 15, 2025) revealed **411,471.7 hectares (4,114.72 km²)** of newly inundated land. The analysis used SAR backscatter thresholding and temporal change detection to accurately map flood extent.

1. INTRODUCTION

1.1 Background

Pakistan experiences recurrent monsoon flooding causing significant agricultural and infrastructure damage. Narowal district in Punjab is vulnerable due to its river proximity and flat terrain. Accurate flood mapping supports disaster response and damage assessment.

1.2 Objectives

- Acquire and process Sentinel-1 SAR imagery for pre/post-flood periods
- Extract water bodies using remote sensing techniques
- Conduct change detection to map flood extent
- Quantify affected area
- Generate professional maps for decision-making

1.3 Study Area

Location: Narowal District, Punjab, Pakistan

Coordinates: ~32.1°N - 32.3°N, 74.8°E - 75.1°E

Terrain: Alluvial plains with river system

2. DATA AND MATERIALS

2.1 Satellite Imagery

Parameter	Specification
Satellite	Sentinel-1A
Sensor	C-band SAR
Product	Ground Range Detected (GRD)
Polarization	VV (Vertical-Vertical)
Resolution	10 meters
Pre-Flood Date	April 29, 2025
Post-Flood Date	September 15, 2025

Source: Copernicus Open Access Hub

2.2 Software

- ArcGIS Desktop / ArcMap / QGIS
- Spatial Analyst Extension

3. METHODOLOGY

3.1 Image Clipping

Purpose: Focus analysis on study area and reduce processing time.

Process:

- Extracted Narowal boundary from Pakistan shapefile
- Used Extract by Mask tool to clip both images

Output: Clipped images (reduced from 800+ MB to manageable size)

Why: Large Sentinel-1 scenes cover excessive area. Clipping improves computational efficiency and focuses on region of interest.

3.2 dB Conversion

Purpose: Convert raw Digital Numbers to calibrated backscatter values.

Process: Applied Raster Calculator formula:

$$\mathbf{dB = 20 \times Log_{10}(DN)}$$

Output:

- Pre-flood: 20.83-83.45 dB
- Post-flood: 20.00-84.54 dB

Why: dB scale standardizes SAR data for consistent analysis. Factor of 20 used because DN represents amplitude (power would use factor of 10). Water appears at low dB values, land at high dB values.

3.3 Water Extraction

Purpose: Classify pixels as water or non-water.

Process: Used threshold of 35 dB in Raster Calculator:

```
Water_Mask = Con(Float(dB_Image) < 35, 1, 0)
```

- Value 1 = Water
- Value 0 = Land

Output: Binary water masks for both dates

Why: Water surfaces are smooth, reflecting radar away (low backscatter < 35 dB). Land is rough, scattering radar back (high backscatter ≥ 35 dB). Threshold of 35 dB accounts for both calm and turbulent floodwater.

3.4 Change Detection

Purpose: Identify newly flooded areas.

Process: Subtracted pre-flood from post-flood water masks:

```
Change_Map = Post_Water - Pre_Water
```

Result Values:

- +1 = New flood (dry → water)

- 0 = No change
- -1 = Water receded (water → dry)

Extracted only flooded areas:

Flood_Map = Con(Change_Map == 1, 1, 0)

Output: Final flood map showing only new inundation

Why: Change detection removes permanent water bodies (rivers, lakes), isolating flood-specific impact on previously dry land.

3.5 Area Calculation

Purpose: Quantify flood extent.

Process:

- Checked attribute table: 41,147,170 pixels with value = 1
- Calculated area: Pixels × (10m × 10m) = 4,114,717,000 m²

Results:

- **411,471.7 Hectares**
- **4,114.72 km²**
- **1,016,698.52 Acres**

Why: Precise quantification enables damage assessment and resource allocation for relief operations.

3.6 Map Creation

Maps Generated:

1. Pre-Flood Map: Baseline conditions (April 29, 2025)

2. Post-Flood Map: During/after flooding (September 15, 2025)

3. Flood Extent Map: Newly inundated areas highlighted

Elements: Title, north arrow, scale bar, legend

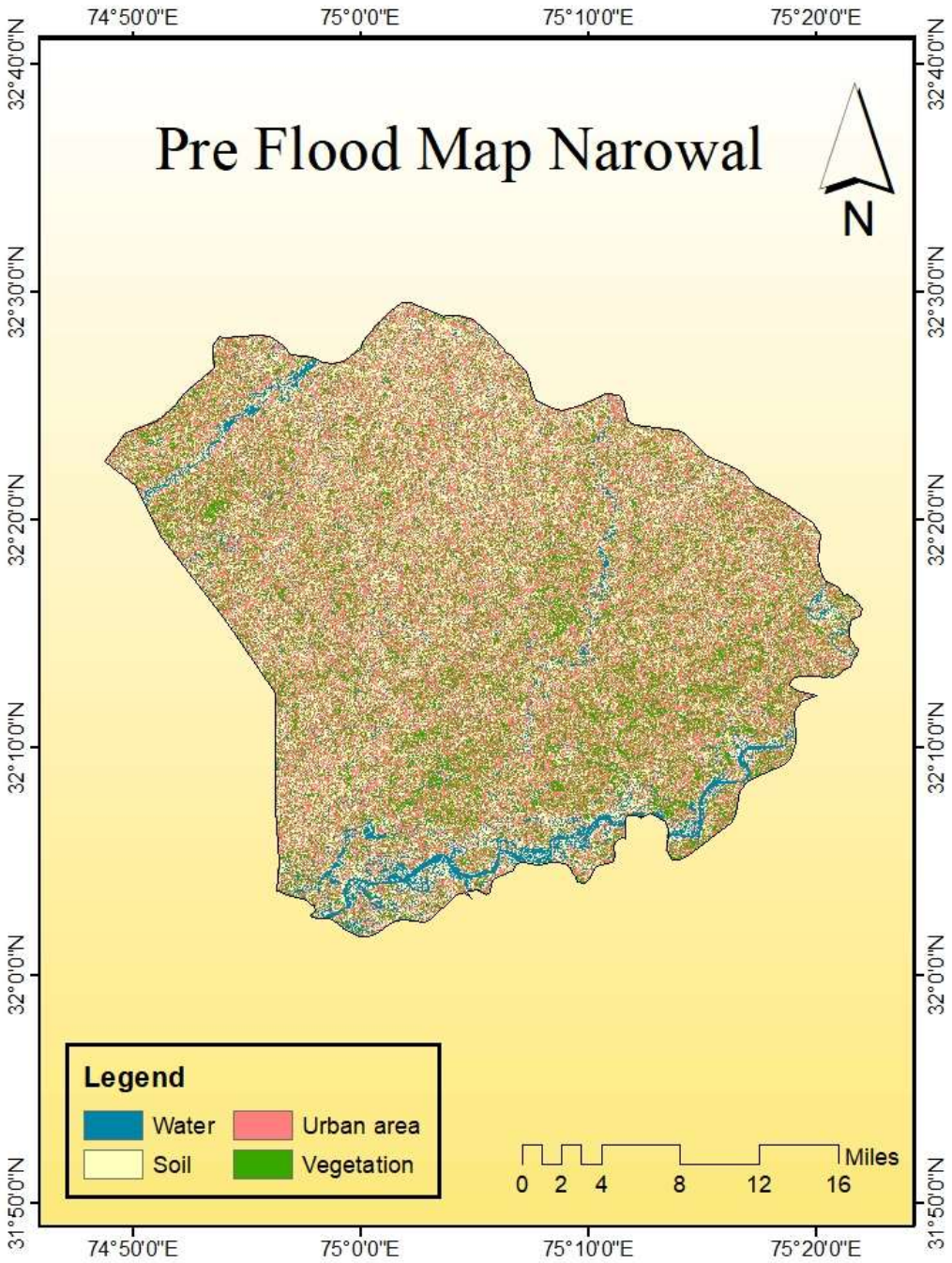
Why: Professional maps communicate spatial information effectively to technical and non-technical stakeholders.

MAPS:

Pre-Flood Map:

Characteristics:

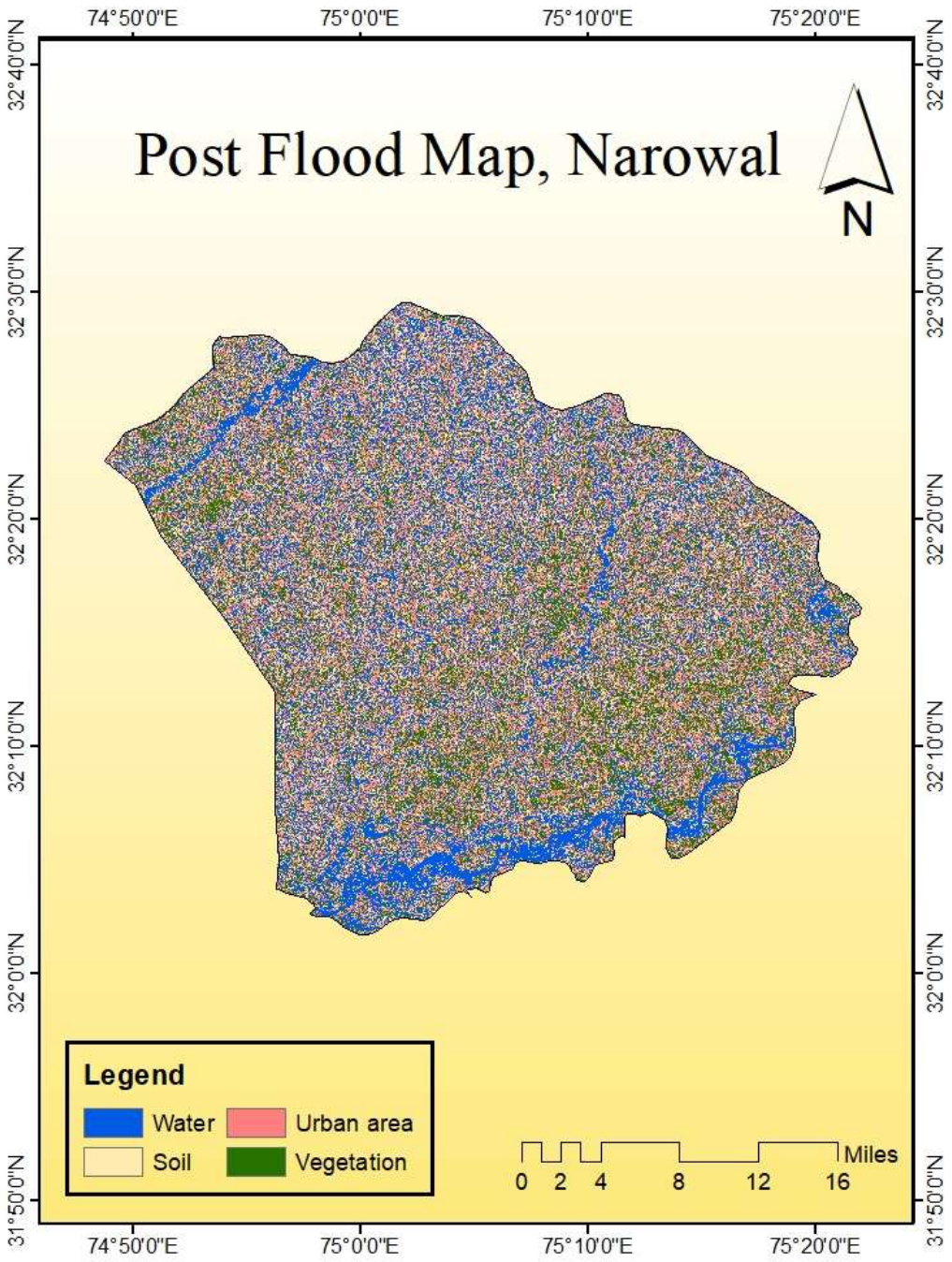
- Predominantly cyan/turquoise color throughout the district
- Minimal blue patches (water)
- Mostly Urban Areas
- Generally uniform appearance



Post-Flood Map:

Characteristics:

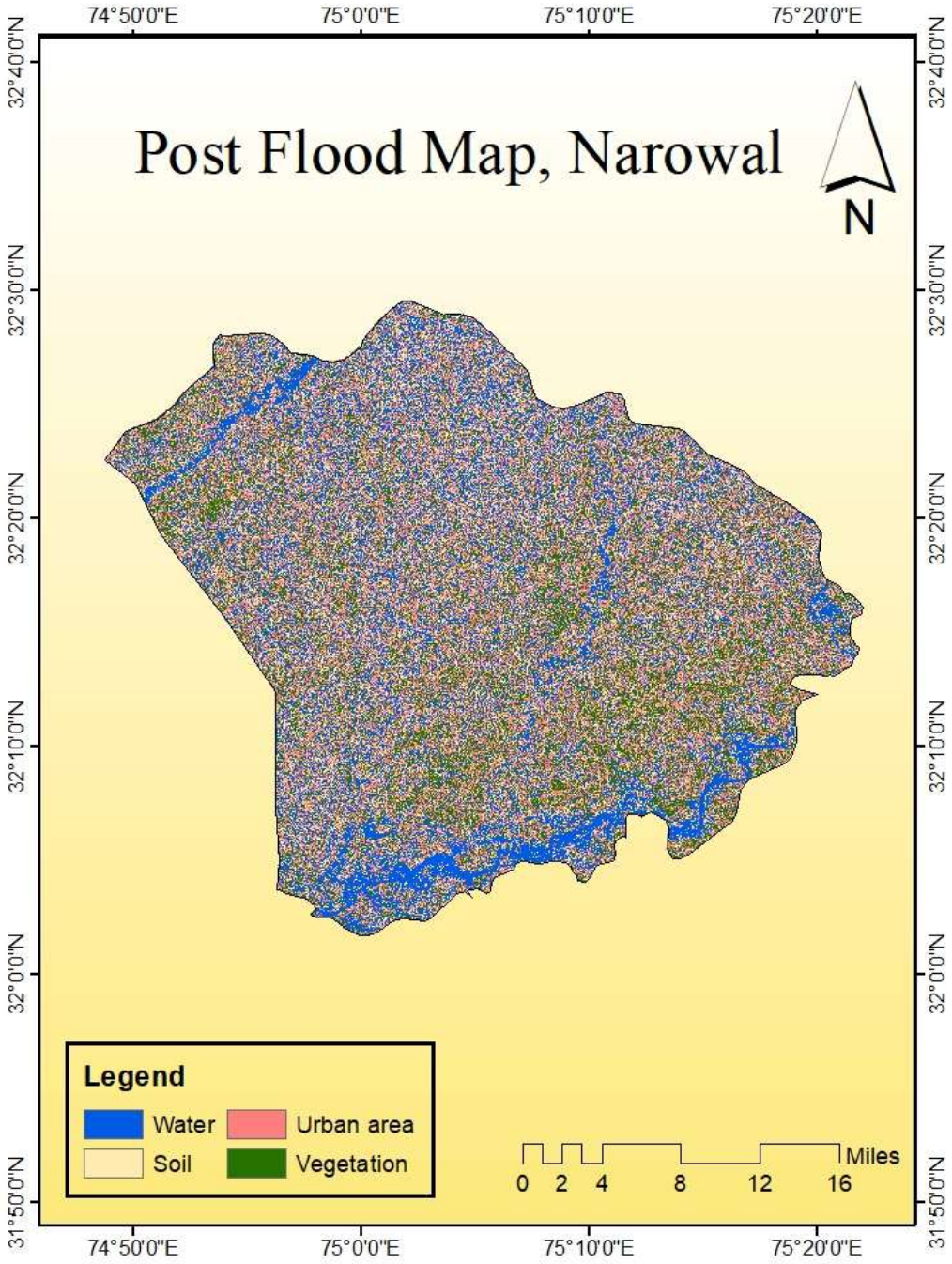
- Mixed blue and green pattern across district
- Significantly more blue areas compared to pre-flood
- Blue distributed throughout, not just in channels
- Striped/linear patterns visible (flood flow patterns)



Final Map of Flood:

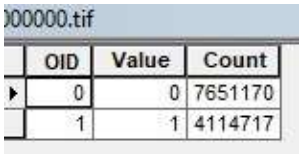
Characteristics:

- Entire district flood is shown by blue print
- Uniform appearance throughout
- Represents the unique flood classification
- Shows comprehensive coverage of the study area



4. RESULTS

4.1 Flood Extent



000000.tif

OID	Value	Count
0	0	7651170
1	1	4114717

Metric	Value
Total Flooded Area	411,471.7 hectares
Equivalent	4,114.72 km² / 1,016,698.52 acres
Affected Pixels	41,147,170

4.2 Spatial Distribution

- Flooding concentrated in low-lying agricultural areas
- Northern and central districts most affected
- Standing water remained in poorly drained zones

4.3 Why Sentinel-1 SAR?

Advantages:

- Cloud penetration: Works through monsoon clouds (optical satellites cannot)
- Day/night capability: Operates 24/7
- Water sensitivity: Strong contrast between water and land
- Free data: No cost, open access
- Regular monitoring: 12-day revisit time

Challenges Addressed:

- Roads can appear similar to water (both smooth) → Solved by change detection
- Turbulent floodwater has higher backscatter → Threshold adjusted to 35 dB

5. CONCLUSIONS

5.1 Key Findings

- 411,471.7 hectares of Narowal flooded between April-September 2025
- Sentinel-1 SAR effectively mapped flood extent despite cloud cover
- Change detection successfully isolated new flooding from permanent water

5.2 Applications

- Disaster response planning
- Damage assessment for compensation
- Future flood risk mapping
- Infrastructure vulnerability analysis

5.3 Recommendations

- Establish regular SAR monitoring during monsoon season
- Integrate with ground surveys for validation
- Develop early warning systems using real-time SAR data
- Create flood susceptibility maps for vulnerable districts

END OF REPORT